

DATA 505: Lab 4

Basic Statistical Inference

YOUR NAME HERE

2025-09-16

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1 Setup & Goals

Just like in Labs 1 and 2, you will edit a `.qmd` Quarto script, render the result to a PDF, and submit both on Moodle (if you do not succeed at rendering the PDF, still submit the `.qmd` script: you may still get full “satisfactory” credit if you have made a reasonable effort).

Goals for Lab 4:

- Use the `t.test()` function to carry out hypothesis tests and to produce confidence intervals for unknown population means.
- Use the `prop.test()` function to do hypothesis tests for unknown population proportions.

2 The data

- The data for this lab come from the `NHANES` package
- This is survey data from the National Health and Nutrition Examination Survey, from 2009-2012

```
# install the MASS package if needed
if(!"NHANES" %in% installed.packages()){
  install.packages("NHANES")
}

# load the NHANES package and verify it is now ready for use
stopifnot(require(NHANES))
```

Loading required package: NHANES

- To load the data from NHANES into your environment, the command is

```
data("NHANES")
```

3 Getting acquainted with the data

- Use the `str()` function to print a concise summary of the `NHANES` object

```
# YOUR CODE HERE
```

- Look up the help page for the dataset to understand more about the columns (recall: to access help in R, you can write in the console `?` followed by the name of the function or object for which you want documentation)

```
?NHANES
```

- We will only work with one of the younger age groups. This code will subset the data to restrict attention to the age 18-25 respondents.

```
sample <- NHANES[NHANES$Age >= 18 & NHANES$Age <= 25,]
```

- Find out how many respondents are left in this subset.

```
# YOUR CODE HERE
```

4 Inference for mean BMI

- Produce and interpret a 95% confidence interval for the mean BMI among 18-25 year olds

```
# YOUR CODE HERE
```

- YOUR INTERPRETATION
- Carry out the following hypothesis test about μ , the mean BMI among 18-25 year olds, and interpret the results.
 - $H_0 : \mu = 27$
 - $H_A : \mu \neq 27$
 - $\alpha = 0.05$

```
# YOUR CODE HERE
```

- YOUR INTERPRETATION

5 Comparing two proportions

- Use the `table()` function to create a two-way frequency table for gender (`Gender`) versus homeownership (`HomeOwn`), but follow the pattern [here in the lecture example](#) to dichotomize homeownership into owners versus non-owners (`HomeOwn == "Own"`).

```
# YOUR CODE HERE
```

- Use `prop.test()` to do a hypothesis test to compare the proportion of homeowners between males and females. Provide a p-value, draw a conclusion, and interpret your results in context.

```
# YOUR CODE HERE
```

- YOUR INTERPRETATION